TEAM ID	SWTID-2026-6767
PROJECT TITLE	Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau
MAX MARKS	5 MARKS

4.2 – Proposed Solution

The proposed solution is to develop an interactive Tableau-based visualization tool that enables users to monitor, analyse, and interpret Electric Vehicle (EV) charging and range data effectively. The tool transforms raw EV datasets into meaningful visual insights, helping users understand battery performance, energy consumption, charging behaviour, and driving range.

The system integrates EV-related data such as battery charge levels, charging duration, distance travelled, energy usage, and charging station information into a centralized Tableau dashboard. Through interactive charts, graphs, maps, and KPIs, users can easily track vehicle performance and identify trends that impact battery efficiency and range.

The solution provides real-time and historical analysis of battery charge status, allowing users to monitor charging patterns and evaluate the effectiveness of different charging strategies. Range prediction features help estimate the remaining travel distance based on battery level and energy consumption patterns, reducing uncertainty during long trips.

Additionally, the dashboard includes geographical visualizations of charging stations, enabling users to locate nearby charging points and plan routes more efficiently. Advanced filtering and drill-down capabilities allow users to analyse data based on vehicle type, trip duration, charging sessions, and other relevant parameters.

By leveraging Tableau's powerful data visualization capabilities, the proposed solution converts complex EV data into intuitive and actionable insights. This helps EV owners, fleet operators, and researchers optimize charging schedules, improve battery utilization, reduce range anxiety, and make data-driven decisions that enhance the overall electric vehicle experience.